

Finance Document Extraction and Reporting

A reusable automation pattern for invoices, transaction documents, reconciliation support, and finance reporting using AI Builder, Azure AI Foundry, Azure Functions, Power Automate, and Power BI.

Sanitized implementation case study

OCR

Document extraction

Reusable

Rule-based parsing

Integrated

Excel / SQL / APIs

Actionable

Variance reporting

What this shows: Finance teams often receive semi-structured documents with different layouts and inconsistent labels. Manual copying slows down processing and makes it difficult to compare reported totals with source documents. The solution pattern separates extraction, normalization, business rules, reconciliation, and reporting so each stage can be tested and improved independently.

End-to-end workflow

1

Invoice / PDF

Source enters the process.

2

OCR + parsing

Context and missing fields are identified.

3

Normalized fields

Data becomes consistent and usable.

4

Business rules

Rules compare, route, or reconcile.

5

Report + action

People receive an action-ready result.

Business value

- Transforms unstructured or incomplete source information into consistent records.
- Separates automation from human review so exceptions are visible.
- Creates a traceable path from original source to report or action.
- Supports operational teams with clearer ownership and faster review.

How to read this project

HR can review the first two pages for the business process and my contribution. The linked repository folder contains sanitized data models, workflow examples, documentation, and implementation notes for deeper technical review.

How the solution works

The important part of this project is the complete operating chain: source capture, structured transformation, business-rule processing, exception handling, and reporting. Each stage produces a clear output that can be validated before the next stage uses it.

01 / SOURCE PROCESSING	Documents are collected from an agreed input location and assigned a batch identifier. The process preserves the original file reference, source date, and processing status for traceability. Output: traceable, reviewable data for the next stage
02 / EXTRACTION	OCR and parsing identify vendor or account information, document number, dates, amounts, taxes, line items, and reference values. Layout-specific patterns and fallback rules handle common variations. Output: traceable, reviewable data for the next stage
03 / NORMALIZATION	Amounts, dates, labels, and identifiers are mapped into a consistent schema. Duplicate checks, missing-field checks, and confidence or exception flags prevent incomplete values from silently entering reports. Output: traceable, reviewable data for the next stage
04 / RECONCILIATION	The normalized records are compared against source reports or operational data. Variances are calculated at the appropriate level, then categorized as missing document, amount mismatch, date mismatch, duplicate, or review required. Output: traceable, reviewable data for the next stage
05 / DELIVERY	Validated records are written to Excel, SQL, Dataverse, or a reporting layer. Power BI-style dashboards can show processed volume, exceptions, aging, variance categories, and operational trends. Output: traceable, reviewable data for the next stage

Example structured record

```
{ "source_reference": "BATCH-2026-001 / DOCUMENT-0042", "record_id": "REC-0042", "status": "review_required", "captured_fields": { "account_number": "sample-account", "document_number": "sample-document", "amount": 0, "tax": 0 }, "checks": { "required_fields": "passed", "duplicate": "not_found", "reconciliation": "variance_found" }, "next_action": "finance_review" }
```

Design principle: Never hide an exception inside a final total. Keep the source reference, extracted value, expected value, variance, and review status together so an operations user can explain what happened.

Data and control points

Control point	Purpose	Typical output
Source reference	Connects the structured row to the physical or digital document.	Batch ID, file ID, page or document reference
Validation	Prevents missing, malformed, duplicate, or low-confidence data from flowing silently.	Pass, fail, review required
Business rules	Applies finance or service logic to compare, route, calculate, or approve.	Matched, variance, assigned queue
Exception record	Makes human review explicit and measurable.	Expected, actual, variance, owner, status
Reporting layer	Turns operational records into trends and actions.	Volume, aging, variance, SLA, resolution

Responsibilities and technology map

My contribution

- Mapped the operational process from source document or user request to final report.
- Defined fields, validation checks, exception states, and review ownership.
- Designed automation steps and structured payloads for repeatable processing.
- Connected data preparation with reporting and operational follow-up.
- Documented the workflow so another team member can understand, test, and operate it.

Portfolio boundaries

This is a sanitized portfolio representation. Casino/client names, real account numbers, production URLs, credentials, and confidential records are intentionally excluded. The domain workflow and technical responsibilities are preserved at a useful level for discussion.

Capability matrix

Capability	Evidence in this project	Keywords
Process automation	Multi-stage intake, transformation, validation, routing, exception, and reporting flow.	Power Automate, workflow design, integration
Data engineering	Structured fields, normalization, staging tables, checks, and downstream reporting data.	Python, SQL, Excel, Power Query, Dataverse
Finance operations	Document capture, transaction fields, reconciliation, variance identification, review states.	OCR, reconciliation, controls, audit trail
Conversational / AI patterns	Where applicable, intent, confidence, fallback, and human-review concepts.	Copilot Studio, Azure AI, Azure OpenAI
Reporting	Operational metrics, variance categories, aging, SLA, resolution and trend views.	Power BI, Microsoft Fabric, dashboards

Technology stack

Python

OCR

AI Builder

Azure AI Foundry

Azure Functions

Power Automate

Excel

Power Query

SQL

Power BI

App Insights

REST APIs

Dataverse

Azure

C# / .NET

JavaScript / TypeScript

Git / CI/CD

Suggested interview walkthrough: Start with the source document or request, explain how you preserved traceability, show the validation and reconciliation decision points, then explain how the final report helped an operations or support user take action.